

Question 1: (a) *Proof.* We induct over n .

BC We compute that when $n = 1$, $f_{4n} = f_4 = f_3 + f_2 = 2 + 1 = 3$, which is a multiple of 3.

IS Assume that f_{4n} is a multiple of 3. Consider $f_{4(n+1)} = f_{4n+4}$. We compute

$$f_{4n+4} = f_{4n+3} + f_{4n+2} = 2 \cdot f_{4n+2} + f_{4n+1} = 3 \cdot f_{4n+1} + f_{4n}.$$

We can see that we $f_{4(n+1)}$ can be expressed as a sum of a multiple of 3 and f_{4n} , which is a multiple of 3 by assumption. Thus, $f_{4(n+1)}$ is also a multiple of 3.

□

(b) *Proof.* We induct over n .

BC We compute that when $n = 1$, $f_{5n} = f_5 = f_4 + f_3 = 5$, which is a multiple of 5.

IS Assume $f_{5n}, n \in \mathbb{N}$ is a multiple of 5. Consider $f_{5(n+1)} = f_{5n+5}$. Similarly to (a), we can rewrite this as:

$$f_{5(n+1)} = f_{5n+5} = 5f_{5n+1} + 3f_{5n}.$$

We can see that that $f_{5(n+1)}$ can be expressed as a sum of a multiple of 5 and a 3 times f_{5n} , which itself is a multiple of 5. Thus, $f_{5(n+1)}$ is a multiple of 5.

Thus, we are done.

□

(c) *Proof.* We induct over n .

BC We compute that when $n = 2$, $\sum_{k=1}^{n-1} f_k = 1 = f_{n+1} - 1$.

IS Assume $f_1 + f_2 + \cdots + f_{n-1} = \sum_{k=1}^{n-1} f_k = f_{n+1} - 1$. Consider $\sum_{k=1}^{(n-1)+1} f_k$.

We then have

$$f_1 + f_2 + \cdots + f_{(n+1)-1} = f_1 + f_2 + \cdots + f_{n-1} + f_n.$$

The sum of the first $n - 1$ terms evaluate to $f_{n+1} - 1$ by assumption. So, we can rewrite as

$$f_{n+1} - 1 + f_n = f_{n+2} - 1.$$

Thus, we are done.

□

(d) *Proof.* We induct over n .

BC We compute that when $n = 1$, $\sum_{k=1}^1 f_{2n-1} = 1 = f_2$.

IS Assume $f_1 + f_3 + \cdots + f_{2n-1} = \sum_{k=1}^n f_{2k-1} = f_{2n}$. Consider $\sum_{k=1}^{n+1} f_{2k-1}$.

Using our assumption, we can rewrite this as

$$\sum_{k=1}^{n+1} f_{2k-1} = \sum_{k=1}^n f_{2k-1} + f_{2n+1} = f_{2n} + f_{2n+1}.$$

We see that $f_{2n} + f_{2n+1} = f_{2(n+1)}$. Thus, we are done. □

(e) *Proof.* We induct over n .

BC We compute that when $n = 1$, $\sum_{k=1}^1 f_{2k} = 1 = f_3 - 1$.

IS Assume that $f_2 + f_4 + \cdots + f_{2n} = \sum_{k=1}^n f_{2k} = f_{2n+1} - 1$. Consider $\sum_{k=1}^{n+1} f_{2k}$.

Using our assumption, we can rewrite this as

$$\sum_{k=1}^{n+1} f_{2k} = \sum_{k=1}^n f_{2k} + f_{2n+2} = f_{2n+1} - 1 + f_{2n+2}.$$

We see that $f_{2n+1} - 1 + f_{2n+2} = f_{2n+3} - 1 = f_{2(n+1)+1} - 1$. Thus, we are done. □

(f) *Proof.* We induct over n .

BC We compute that when $n = 1$, $\sum_{k=1}^1 f_k^2 = 1 = f_1 f_{n+1}$.

IS Assume that $f_1^2 + f_2^2 + \cdots + f_n^2 = \sum_{k=1}^n f_k^2 = f_n f_{n+1}$. Consider $\sum_{k=1}^{n+1} f_k^2$.

Using our assumption, we can rewrite this as

$$\sum_{k=1}^{n+1} f_k^2 = \sum_{k=1}^n f_k^2 + f_{n+1}^2 = f_n f_{n+1} + f_{n+1}^2.$$

We can rewrite this as $f_{n+1}(f_n + f_{n+1}) = f_{n+1} f_{n+2}$. Thus, we are done. □

(g) *Proof.* We induct over n .

BC We compute that when $n = 1$, $f_1 = 1$, which is odd.

IS Assume $n \not\equiv 0 \pmod{3}$, and f_n is odd. If $n \not\equiv 0 \pmod{3}$, there are two cases: $n \equiv 1 \pmod{3}$ or $n \equiv 2 \pmod{3}$. We need not consider both cases; we know that for any n such that $n \not\equiv 0 \pmod{3}$, $n + 3 \not\equiv 0 \pmod{3}$. WLOG, consider f_{n+3} . We can rewrite this as

$$f_{n+3} = f_{n+2} + f_{n+1} = 2 \cdot f_{n+1} + f_n.$$

By assumption, f_n is odd, and thus f_{n+3} is the sum of an even and an odd, which is always odd.

Thus, we are done. □

Question 2: (a) Using the Euclidean Algorithm, we find that $\gcd(775, 682) = 31 = 8 \cdot 682 - 7 \cdot 775$.

(b) Similarly, we find that $\gcd(562, 243) = 1 = 16 \cdot 562 - 37 \cdot 243$.

Question 3: *Proof.* We show that $S = d\mathbb{Z}$ by showing set equality.

- ($\subseteq$) Let m be an element of $d\mathbb{Z}$. So, $m = kd, k \in \mathbb{Z}$. By Bézout's Identity, $\gcd(a, b) = d$ can be expressed as $ax + by = d$ for some $x, y \in \mathbb{Z}$. So, $m = k(ax + by) = (ka)x + (kb)y$. $ka, kb \in \mathbb{Z}$. Thus, $m \in S$, and $d\mathbb{Z} \subseteq S$.
- ($\supseteq$) Let n be an element of S . Thus, $n = ax + by$ for some $x, y \in S$. Let $d = \gcd(a, b)$. By definition, $d \mid a$ and $d \mid b$. Thus, there exist $k, k' \in \mathbb{Z}$ such that $a = dk$ and $b = dk'$. Subbing in, we get that $n = (dk)x + (dk')y = d(kx + yk')$. Thus, $n \in d\mathbb{Z}$. So, $S \subseteq d\mathbb{Z}$.

Thus, we are done. $\square$

Question 4: *Proof.* By the Bézout Identity, there exist $x, y, x', y' \in \mathbb{Z}$ such that $ax + cy = 1$ and $bx' + cy' = 1$. We multiply these equations against each other to obtain:

$$(ax + cy)(bx' + cy') = abxx' + axcy' + cybx' + c^2yy' = \\ ab(xx') + c(axy' + ybx' + cyy') = 1.$$

Every quantity is an integer, and thus we have a linear combination of ab and c that sums to 1. Thus, $\gcd(ab, c) = 1$. $\square$

Question 5: The error is assuming that a^{n-1} actually exists. There is no guarantee that the value of n is not less than or equal to 0.

Question 6: We can see that the Well-Ordering Principle and induction are closely related by thinking about the set of all the integers that make the open sentence $P(n)$ false. If we have k as the least element, then $P(k - 1)$ must be true. Then, using the inductive step, we see that $P(k)$ must be true, but k was an element of the set of the integers that make $P(n)$ false, so we have a contradiction. Using this method, the base case and inductive step are sort of like holding a car whose tires are already spinning above the road, and then dropping it, and then it zooms off.